



Cambridge IGCSE™

PHYSICS

0625/42

Paper 4 Extended Theory

February/March 2026

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **17** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	evaluation attempted
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response has not answered question

Annotation	Meaning
RE	rounding error
SEEN	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
TE	transcription error
TV	response is too vague or there is insufficient detail in response
T	answer outside the tolerance of the mark scheme
	used to highlight parts of an extended response
	used to highlight parts of an extended response
MO	mandatory mark not awarded
SC	special case

Specific Instructions for Marking 0625 / Paper 4

Preparation for Marking

Instructions and handbooks for markers using RM Assessor 3 can be found at [RM support portal](#).

Marking

M1. Blank pages, additional objects and marking outside the question zone.

Blank pages will be attached to the first part of **Q1** and should be annotated with SEEN on all scripts.

Annotate any blank Additional Objects with SEEN.

Link any other additional objects to the question or questions applicable.

Examiners must ensure that they view the whole exam paper for each candidate. This will sometimes mean scrolling through a large zone to ensure that no working relevant to either the current or any other question is missed.

Where a candidate's answer extends beyond the marking zone, examiners must view the whole page (or link to other pages) to annotate and mark the whole answer. To view the whole page, deselect any annotation tool from the mouse, then click in the bottom right-hand corner of the marking zone where "view whole page" appears. For instructions to link to other pages see above.

M2. Use of Annotation stamps.

Examiners annotate scripts to explain their reasons for awarding or not awarding marks, noting:

- for **all** questions with **two** or **more** marks, it is **mandatory** to annotate with ticks placed to indicate where each mark is awarded. In a calculation where the final answer (A) mark is awarded all the ticks should be placed near to the final answer.
- annotations and comments must never suggest or imply that a mark has been deducted e.g. –1
- annotation stamps and their uses are published with the mark scheme and visible on scripts returned to centres. It is therefore vital that examiners apply them to scripts in a manner consistent with their published meaning.

M3. Acronyms and shorthand in the mark scheme.

acronym/shorthand	explanation
A mark	Final answer mark which is awarded for fully correct final answers
C mark	Compensatory mark which may be scored when the final answer (A) mark for a question has not been awarded.
B mark	Independent mark which does not depend on any other mark.
M mark	Method mark which must be scored before any subsequent final answer (A) mark can be scored.
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word/phrase/unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
ORA	Or reverse argument
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	an answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
cao	correct answer only

M4. Miscellaneous**Equations and formulae.**

- Where a C, B or M mark is available for quoting a formula or equation this can be done in any form and in words, symbols or numbers unless the mark scheme specifies otherwise.
- Where a C mark is available for quoting an equation and another C mark is available for rearranging and/or substituting numbers into an equation, a candidate who only writes down the equation in the rearranged (and/or numerical) form in one step gains both C marks.

Use of ecf. The mark scheme notes where ecf is applicable, in the guidance section of the final answer mark. However, it should be applied for all relevant C marks as well. **Always annotate ecf if applied.** See Science specific Marking point 4 above.

Units.

- Where the unit for a numerical answer is not included on the question paper and if the mark scheme doesn't specify otherwise, the only permitted derived units are:

unit	permitted derived units
W	J / s or N m / s
Pa	N / m ²
momentum	N s or kg m / s
impulse	N s or kg m / s
J	N m

- Note: J is **not** permitted as the unit for moments.

Significant Figures.

- Numerical answers are expected to be given to the number of significant figures given in the final answer in the mark scheme.
- Answers given to **more** SFs than expected are awarded the A mark if, when rounded to the correct number of SF, the answer matches the mark scheme answer (allowing for ECF).
- Where a final answer is given to **too few** SF, and the mark scheme answer **has been seen** in the working to at least the expected number of SF, this is penalised once only on the whole paper, with SF annotation and A mark is not awarded. Subsequent occurrences of this are ignored and the A mark is awarded.
- Where a final answer is given to **too few** SF, and the mark scheme answer **has not been seen** in the working to at least the expected number of SF, this is treated as an arithmetic error, and the A mark is not awarded.

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Arithmetic errors and Transcription errors. Where a (probable) transcription error or arithmetic error is identified in working, ignore the error when awarding C marks.

Fractions. A final answer expressed as a fraction is only accepted if explicitly stated in the mark scheme.

Crossed out work. When only part of an answer is crossed out the crossed-out work must be ignored. However, work which has been **wholly** crossed out and not replaced and can easily be read, should be marked as if it had not been crossed out. Look to see if it has been replaced on a blank page or another part of the same page before attempting to mark the crossed-out work.

Marking diagrams on-screen. Differences in magnification and/or individual computer screen settings can alter the appearance of diagrams. If it is necessary to check line lengths or angles use the ruler and protractor tools provided within RM Assessor 3 to ensure consistency across all examiners.

NR. (# or / key on the keyboard). Use this (instead of giving 0 marks) if the answer space for a question is completely blank or contains no readable words, figures or symbols.

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Question	Answer	Marks
1(a)	1.7 (km)	B1
1(b)	B	B1
	C OR D	B1
1(c)	20 (km / h) OR maximum speed, correctly calculated from the gradient of BC using values other than point B and point C.	A3
	(speed =) gradient (of distance-time graph) OR triangle method for gradient shown on graph OR (Speed =) $\{1.25 - 0.4 \div \{7.5 - 5.0\}\}$ OR (Speed =) $0.85 / 2.5$ OR 0.34	C1
	conversion from km / min to km / h OR conversion from min to h	C1

Question	Answer	Marks
2(a)(i)	$\rho = m / V$ OR density equals mass divided by volume	B1
2(a)(ii)	0.014 (m ³)	B1
2(a)(iii)	Any one from: - steel is denser (than sea water) - density (sea) water < density of steel 1030 (kg / m³) < 7900 (kg / m³) - it has higher density (than sea water)	B1

Question	Answer	Marks
2(a)(iv)	Any one from: - ship is not solid (steel) - the volume of the ship includes air - air is (a lot) less dense than (sea) water - air reduces density of the ship	B1
	Any one from: - density of the ship < density of (sea) water - it is less dense than (sea) water - average density (of steel + air) < density of (sea) water	B1
2(b)(i)	2.6×10^7 (N)	A2
	$(F=) ma$ OR $(F=) 3.5 \times 10^7 \times 0.75$	C1
2(b)(ii)	Any two from: (there is a) resistive force OR named resistive force (resistive) force acts on the ship OR force opposes (the engine) - force in (b)(i) is the resultant force OR resultant force is less (than the force from the engine)	B2

Question	Answer	Marks
3(a)(i)	5800 (N)	B1
3(a)(ii)	5200 (Pa)	A2
	$(P=) F / A$ OR $5800 \div 1.12$	C1
3(a)(iii)	7(.0) AND 56	B1
	8(.0)	B1

Question	Answer	Marks
3(b)	25 000 (Pa) OR 2.5×10^4 (Pa)	A2
	$(\Delta) = \rho g(\Delta)h$ OR $(\Delta p =) 1030 \times 9.8 \times 2.50$	C1

Question	Answer	Marks
4(a)	radiation	B1
4(b)(i)	thermal OR internal	B1
	(a word meaning) absorbed OR gained OR taken in (a word meaning) emitted OR lost OR released	B1
4(b)(ii)	(thermal) energy absorbed (by the table) decreases OR (rate of) emission from table is greater than (rate of) absorption (by table)	B1
	(because) white is a good reflector of radiation OR (because) white is a poor absorber of radiation	B1
4(c)	7.5 (kWh)	A 2
	$(E =) Pt$ OR $(E =) 2.5 \times 3$	C1

Question	Answer	Marks
5(a)	0.39 (m ³)	A3
	PV= constant	C1
	$(V =) \{1.02 \times 1(.0) \times 10^5\} \div 2.6 \times 10^5$ OR $(V =) 3.9 \times 10^N$	C1

Question	Answer	Marks
5(b)(i)	(separation) large OR (particles) far apart	B1
	(arrangement) irregular OR random	B1
	(motion) rapid OR random (directions)	B1
5(b)(ii)	<u>average</u> kinetic energy (KE) of <u>particles</u> decreases	A2
	kinetic energy (KE) of <u>particles</u> decreases	C1

Question	Answer	Marks										
6(a)(i)	principal axis	B1										
6(a)(ii)	ray parallel to principal axis between tip of <i>O</i> or <i>I</i> and lens AND ray from intersection of this ray with centre of lens to the tip of <i>I</i> or <i>O</i>	B1										
	1.8 (cm)	B1										
6(a)(iii)	real	B1										
	same size	B1										
6(b)(i)	X between <i>F</i> and the lens.	B1										
6(b)(ii)	<table><tr><td>on the opposite side of the lens to the object</td><td></td></tr><tr><td>on the same side of the lens as the object</td><td>✓</td></tr><tr><td>the same distance from the lens as the object</td><td></td></tr><tr><td>closer to the lens than the object</td><td></td></tr><tr><td>further away from the lens than the object</td><td>✓</td></tr></table>	on the opposite side of the lens to the object		on the same side of the lens as the object	✓	the same distance from the lens as the object		closer to the lens than the object		further away from the lens than the object	✓	B1
	on the opposite side of the lens to the object											
	on the same side of the lens as the object	✓										
	the same distance from the lens as the object											
	closer to the lens than the object											
further away from the lens than the object	✓											

Question	Answer	Marks
7(a)	4.5	B1
	V	B1
7(b)	charge AND time	B1
7(c)	1500 (Ω)	A3
	$(R =) V \div I$ OR $(R =) 9(.0) \div 4.6 \times 10^{-3}$ OR $(V_R =) 4.6 \times 10^{-3} \times 480$	C1
	(resistance of thermistor =) total resistance – 480 OR (share of e.m.f. for thermistor =) $9(.0) - 2.2$ OR $(R =) \{9 - \{4.6 \times 10^{-3} \times 480\}\} \div 4.6 \times 10^{-3}$	C1
7(d)	(voltmeter) unchanged AND (ammeter) decreases	B1

Question	Answer	Marks
8(a)(i)	33 000 (V) OR 3.3×10^4 (V)	A2
	$(V_p =) N_p V_{s1} \div N_s$ OR $(V_p =) \{6200 \times 230\} \div 43$	C1
8(a)(ii)	0.067 (A) OR 0.066 (A)	A3
	$(I_p =) I_s V_s \div V_p$ OR $(I_p =) 2200 \div 33\,000$	C1
	(conversion 2.2 kW to) 2200 (W) OR $(I_p =) 6.6 \times 10^N$ OR $(I_p =) 6.7 \times 10^N$	C1
8(b)(i)	$P = I^2 R$	B1

Question	Answer	Marks
8(b)(ii)	high voltage reduces current (for same power) OR high voltage means low current (for same power)	B1
	(electrical) power (output) is the same (with lower current) OR	B1
	(low current gives) less heating effect ORA	

Question	Answer	Marks
9(a)	path labelled γ undeviated through the electric field	B1
	path labelled α is a <u>curve</u> towards <u>negative</u> plate AND deflects when it enters the space between the plates	A2
	path labelled α is a straight line towards <u>negative</u> plate AND deflects when it enters the space between the plates	C1
	path labelled α continues horizontal between the plates and then <u>curves</u> towards <u>negative</u> plate	C1
9(b)	(α -particles' path is) stopped (by paper) OR α -particles don't pass through (electric field)	B1
	(γ -radiation path) is unchanged	B1
9(c)	(statement) β -particles deflected (towards positive plate) OR (path) curves (upwards)	B1
	(explanation: β -particles) are (negatively) charged	B1
	charged particles in an electric field experience a force owtte	B1

Question	Answer	Marks
10(a)(i)	red giant	B1
10(a)(ii)	white dwarf (star)	B1

Question	Answer	Marks
10(b)(i)	blue light has a shorter wavelength (than red light) OR it has a shorter wavelength ORA	B1
10(b)(ii)	6.0×10^{14} (Hz)	A2
	$(f =) c / \lambda$ OR $(f =) 3.0 \times 10^8 \div 5.0 \times 10^{-7}$	C1

Question	Answer	Marks
11(a)	(r is) radius of <u>orbit</u> (of the object)	B1
	(T is) orbital period OR time taken to (complete one) orbit (around the Sun)	B1
11(b)(i)	Any two from: - orbit is not (exactly) circular OR orbit is (only) approximately circular - r is an average value - r varies OR distance from the Sun varies A Sun's gravitational pull varies	B2
11(b)(ii)	$(T_{\text{Saturn}} =) 9.16 \times 10^8$ (s) OR $(T_{\text{Saturn}} =) (2\pi \times 1.4 \times 10^{12}) / (9600)$	A2
	$(T =) 2\pi r / v$	C1
	$(T_{\text{Earth}} =) 3.15 \times 10^7$ (s) OR $(T_{\text{earth}} =) 365 \times 24 \times 60 \times 60$ (s)	B1
	$((T_{\text{saturn}} =) 29$ (Earth) years	B1